

Part One: Foundations, Regression, and Generalized Linear Models

An Introduction to the Chapters and How They Fit Together

Jinsong Chen

Table of contents

1	What Part One Offers	1
2	How the Chapters Build on Each Other	2
3	How to Read a Chapter and Use the Concept Reference	5
4	What Comes Later	6

1 What Part One Offers

Quantitative Methods for Social and Behavioral Research develops the statistical understanding needed to read empirical studies, interpret their results, and conduct quantitative research. Part One is where the book begins: fourteen numbered chapters that start with what a study asks and how it produces its evidence, and end with regression models for a range of outcomes. It is written for postgraduate students and applied researchers. A mean, an interval, or a regression coefficient becomes useful when a reader understands what it describes, how it was obtained, and which question it helps answer.

Some introductory familiarity with statistics and algebra is useful, and little else is assumed. Each chapter explains the notation it uses and develops its calculations in worked steps, so the algebra arrives with the idea that needs it. Calculus and proof-based mathematical statistics are not prerequisites. Chapters 1–4 can be read without running R or SPSS: Chapter 1 introduces the research journey, and Chapters 2–4 teach statistics directly, through explanations, figures, and worked calculations. Computation joins the statistics in Chapters 5 and 6.

Results are rounded to four decimals from full-precision computation, with trailing zeros dropped (3.1 rather than 3.1000), unless a source’s own precision applies.

2 How the Chapters Build on Each Other

Nine chapters are available, and they form one route. Chapter 1, *The Quantitative Research Journey*, orients the reader in questions, evidence, design, and interpretation. Chapter 2, *Data, Distributions, and Variation*, develops records, distributions, and variation across repeated samples. Chapter 3, *Estimates, Uncertainty, and Statistical Tests*, develops estimation, intervals, and tests. Chapter 4, *Group Comparisons with Analysis of Variance*, develops comparisons among several groups. Chapter 5, *Description and Basic Inference with R*, and Chapter 6, *Group Comparisons and Statistical Modelling with R*, make those foundations executable. Chapters 7–9 then develop regression from one predictor to the regression toolbox: Chapter 7, *Simple Linear Regression*; Chapter 8, *Multiple Regression I: Conditioning, Checks, and Coordinated Questions*; and Chapter 9, *Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty*.

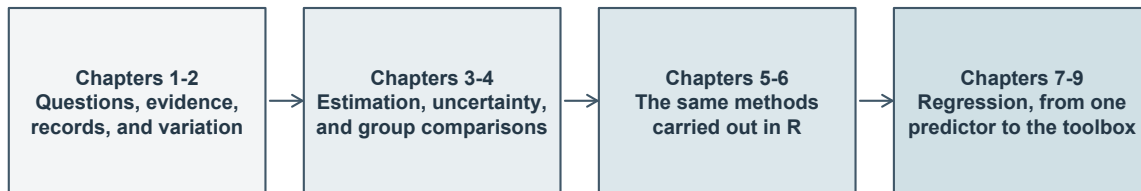


Figure 1: The route through the available chapters, from questions and records to inference, computation, and regression.

Table 1 states what each chapter is responsible for and what it builds on.

Table 1

What each available chapter is responsible for.

Chapter	What it is responsible for	What it builds on
1. The Quantitative Research Journey	Turning a concern into a question with a stated target quantity; how a study produces its evidence; how comparison, timing, sampling, and assignment support descriptive, associational, predictive, and causal answers; reading a reported result as a bounded claim	The starting point of the part

Chapter	What it is responsible for	What it builds on
2. Data, Distributions, and Variation	Reading records, identifiers, codes, and denominators; sampling schemes; distributions, their skew and tail weight, centre, spread, and groups; relationships between two variables; how a summary varies across repeated samples	Reads the dataset behind Chapter 1's account of evidence and applies its design reasoning to an interpretation
3. Estimates, Uncertainty, and Statistical Tests	The target quantity, the rule that estimates it, and the value obtained; standard errors for a mean and a difference, including the pooled standard error, and the conditions behind them; confidence intervals and confidence level; hypothesis tests, p-values, and significance; the one-sample, paired, equal-variance, and Welch t tests; the chi-square test of association and tables of three or more categorical variables; magnitude, error rates, power, and choices made after seeing the results	Turns Chapter 2's sampling distributions and standard error into inference from one observed sample
4. Group Comparisons with Analysis of Variance	Group-mean models and their comparison by the F test; one-way and two-way analysis of variance; sums of squares, degrees of freedom, and mean squares; contrasts and comparison families; interaction and simple effects, extended to a worked three-way table and the structure of four-way designs	Applies Chapter 2's description of group variation and Chapter 3's inference to comparisons among several groups
5. Description and Basic Inference with R	Preparing records in base R with explicit missing values; distributions, graphics and probability calculations; relationships, covariance and correlation matrices, and one-control partial correlation; simulated sampling variation, bias and standard error; mean intervals and their repeated-study coverage, z and t tests, and selected correlation inference; a short reproducible report	The computational route through Chapter 2's order of distributions, relationships and sampling, followed by Chapter 3's mean inference

Chapter	What it is responsible for	What it builds on
6. Group Comparisons and Statistical Modelling with R	Group-mean and factorial models in R, with the F test comparing named models; graphical comparisons with ggplot2; contrasts and comparison families, including a declared contrast and the protection needed if it had been selected after inspection; proportions and counts, with their effect scales; numerical optimization through squared loss and a scalar likelihood; simulation, precision and power	Implements Chapter 4's model comparisons and extends Chapter 3's categorical and planning lessons on Chapter 5's computational foundation
7. Simple Linear Regression	The model for a conditional mean and its error variation, first through a known process and then occupation records; least-squares estimates and slope inference; repeated-study uncertainty; mean-response and new-case intervals; diagnostics; regression toward the mean, its paradox and its causal misreading, with Galton's heights and repeated exams	Extends Chapter 4's group-mean model to a line, using Chapters 5–6's covariance, inference, simulation and objective-function work
8. Multiple Regression I: Conditioning, Checks, and Coordinated Questions	The multiple-regression model and partial slopes through a known process; estimation and inference on the same occupation records, with the overall F test; checks, coefficient covariance, fitted profiles and standardization; a separate held-out prediction question; three coordinated answers in one study	Extends Chapter 7's model and line with Chapter 5's partial correlation and Chapter 4's model comparisons, implemented in R in Chapter 6
9. Multiple Regression II: Categorical Predictors, Model Comparison, Nonlinearity, and Robust Uncertainty	Changes to the mean part: indicators and contrasts, covariance adjustment, nested and non-nested model comparisons, and changed forms and scales; changes to uncertainty: HC3, resampling, and a randomized study with school clusters	Extends Chapter 8's model, connecting each method to the part it changes and the conditions it needs

Each idea has one principal home, and later chapters use it rather than teach it again. Chapter 1 previews uncertainty while reading a study; Chapter 2 develops sampling distributions and

the standard error through repeated samples; Chapter 3 develops their inferential use and states the conditions they require. Chapter 2 explains the descriptive ingredients of group variation; Chapter 4 develops the full analysis-of-variance decomposition and its inferential use, comparing one common mean with group means and an additive model with a model that includes interaction. Where a later chapter needs an earlier idea, it recalls it briefly and then puts it to work in the new setting.

3 How to Read a Chapter and Use the Concept Reference

Chapters 1–9 share an arrangement. A **Chapter overview** box opens the chapter with the route it follows and what a reader will be able to do afterwards. Numbered sections develop the material in order. A closing **Reading Bridge** brings the principal ideas together and connects them to the next chapter, and a selective **Key terms** table gives a compact reference to the concepts explained principally there. Unnumbered material follows the numbered sections: exercises for working with the ideas, appendices holding fuller derivations and specialized extensions, and the references. Later chapters may use a different arrangement as the collection grows.

There are two sensible routes through the available chapters. The first is the chapter sequence. The second pairs statistics with computation: Chapters 2 and 3 with Chapter 5, then Chapters 3 and 4 with Chapter 6. Either way, follow a method from its data and question through its calculation to its interpretation; a result separated from the question it answers is difficult to use.

The **Part One Concept Reference** is published beside this introduction. It gathers selected concepts from Chapters 1–9 into six tables, arranged by concept group and then by term, with each term’s meaning and the place where the book explains it principally. A further table lists the procedures of Chapters 3 and 4 with what each estimates and the conditions it relies on, and a closing table records the notation those chapters use. It is a place to look something up while reading rather than a substitute for reading, because first-use explanations and worked examples stay in the chapters themselves.

Readers who would like to run R without installing anything can use the public browser practice pages that accompany Chapters 3 and 4.

4 What Comes Later

Part One continues with five further chapters. Chapter 10, *Interaction and Moderation*, develops product terms, conditional slopes, and probing within joint support. Chapter 11, *Binary Logistic Regression*, develops the Bernoulli likelihood, the logit link, and the probability, odds, and log-odds scales. Chapter 12, *Extending the GLM: Nominal, Ordinal, and Count Outcomes*, varies only the response distribution and the link for nominal, ordered, and count or rate outcomes. Chapter 13, *Prediction, Cross-Validation, and Regularization*, develops prediction as a separate aim, with held-out error and penalized fitting. Chapter 14, *Regression Pitfalls and Causal Design*, examines what can undermine a regression-based claim: reversals and Simpson’s paradox, regression to the mean, selection, measurement, dependence, and what a study’s design supplies that adjustment cannot.

Three argument chapters extend the argument-based approach. Argument Chapter A, *How to Read a Quantitative Study as an Argument*, develops the reading, evaluation, and construction of one question-level argument. Argument Chapter B, *Extending the Argument-Based Approach to Quantitative Studies*, develops arguments with several statistical strands and studies with coordinated questions. Argument Chapter C concerns arguments involving latent variables or an additional measurement layer, drawing on Part Two. The argument chapters are optional, and no numbered chapter depends on them.

Chapters 10–14 and Argument Chapters A, B, and C are not yet available.

Part Two of the book turns to latent-variable modeling, where the variable of primary interest is one that no column of the data set holds.